

QUESTION ONE

(a) Evaluate:

(i) $\int_0^1 \frac{x}{x^2 + 1} dx,$

(ii) $\int_{-2}^{2\sqrt{3}} \frac{1}{4 + x^2} dx.$

(b) Find the gradient of the tangent to the curve $y = \tan^{-1}(\sin x)$ at $x = 0$.

(c) Solve $\frac{1}{x+1} < 3$.

(d) Give the general solution of the equation $\cos(\theta + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}$.

(e) If $f(x) = 8x^3$ then find the inverse function $f^{-1}(x)$.

QUESTION TWO

(a) Prove the identity $\frac{\sin 2x}{1 + \cos 2x} = \tan x$.

(b) The equation $x^3 - 103x + 102.5 = 0$ has a root near $x = 1$. Take $x = 1$ as a first approximation and use Newton's method once to obtain a closer approximation to this root.

(c) (i) Sketch the graph of $y = |2x - 4|$.

(ii) Using your graph or otherwise solve the inequation $|2x - 4| > x$.

(d) (i) Express $7 \cos \theta - \sin \theta$ in the form $R \cos(\theta + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$.

(ii) Hence solve $7 \cos \theta - \sin \theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$, giving your answer to the nearest degree.

QUESTION THREE

(a) Consider the function $f(x) = 3 \sin^{-1}(\frac{x}{2} - 1)$.

(i) State the domain of $f(x)$.

(ii) State the range of $f(x)$.

(iii) Sketch the graph of $y = f(x)$.

(iv) Evaluate $f(1)$.

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(b) Find $\int x\sqrt{1-x} dx$, using the substitution $u = 1 - x$.

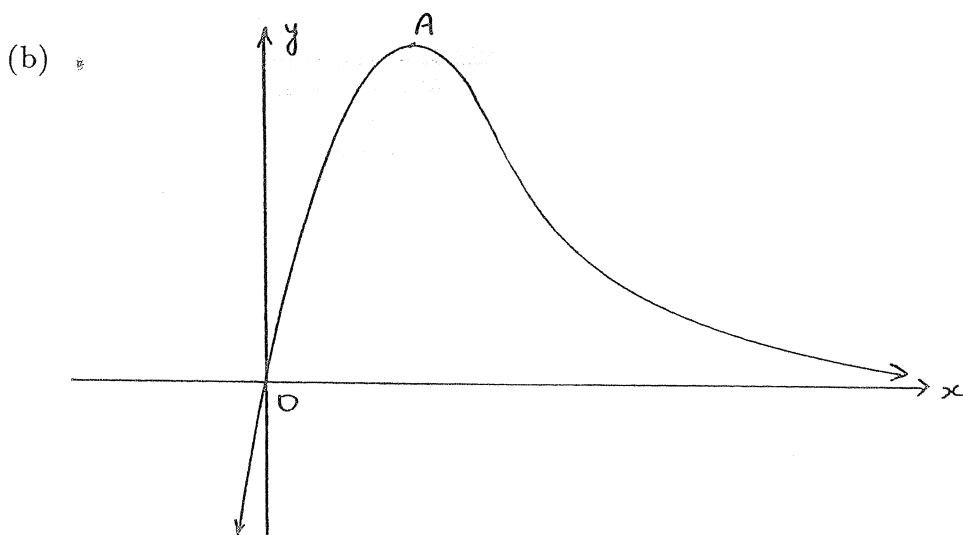
(c) Find the values of the constants a and b if $x^2 - 2x - 3$ is a factor of the polynomial $P(x) = x^3 - 3x^2 + ax + b$.

(d) (i) If $x > 0$, prove that $\frac{d}{dx} (\tan^{-1} x + \tan^{-1} \frac{1}{x}) = 0$.

(ii) Hence find the value of $\tan^{-1} x + \tan^{-1} \frac{1}{x}$ for $x > 0$.

QUESTION FOUR

(a) Find $\cos \theta$ if $\theta = \cos^{-1} \frac{24}{25} - \sin^{-1} \frac{15}{17}$.



The diagram shows the graph of the function $y = xe^{-x}$. A is a stationary point on the curve.

(i) Show that A is the point $(1, \frac{1}{e})$.

(ii) State the range of the function $y = xe^{-x}$.

(iii) How many real roots are there to the equation $xe^{-x} = k$ if:

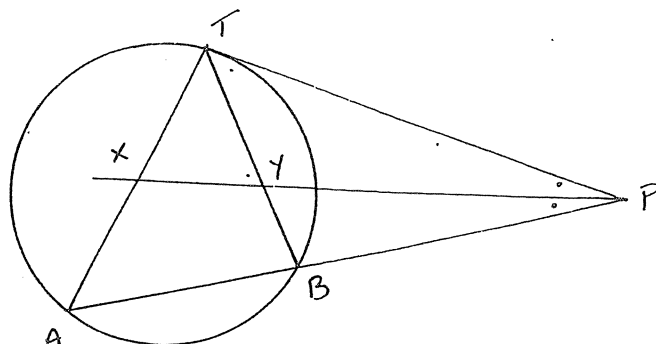
(α) $0 < k < \frac{1}{e}$,

(β) $k \leq 0$,

(γ) $k > \frac{1}{e}$?

(Exam continues overleaf ...)

(c)



The tangent at T on the circle meets a chord AB produced to P . The bisector of $\angle TPA$ meets TA and TB at X and Y respectively.

- (i) Give the reason why $\angle PTB = \angle TAB$.
- (ii) Prove $TX = TY$.
- (iii) Prove $\frac{TX}{XA} = \frac{TP}{PA}$.

QUESTION FIVE

- (a) A particle P moves along the x -axis so that at time t seconds it is x cm from the origin O and its velocity is v cm/s. Initially the particle is at rest at the origin.
 - (i) If the acceleration of P is given by $\ddot{x} = 4(40 - x)$ cm/s², use $\ddot{x} = \frac{d}{dx}(\frac{1}{2}v^2)$ to show $v^2 = 4(80x - x^2)$.
 - (ii) Prove that P moves in the interval $0 \leq x \leq 80$.
 - (iii) Find the maximum velocity of the particle and where the maximum occurs.
- (b) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$.
 - (i) Show that the equation of the normal to the parabola at the point P is $x + py = 2ap + ap^3$.
 - (ii) If the normal at P cuts the y -axis at Q , show that the co-ordinates of Q are $(0, 2a + ap^2)$.
 - (iii) Show that the co-ordinates of R which divides the interval PQ externally in the ratio $2 : 1$ are $(-2ap, 4a + ap^2)$.
 - (iv) Find the Cartesian equation of the locus of R and describe this locus in geometric terms.
 - (v) Show that if the normal at P passes through a given point (h, k) then p must be a root to the equation $ap^3 + (2a - k)p - h = 0$.
 - (vi) What is the maximum number of normals of the parabola $x^2 = 4ay$ which can pass through any given point? Give reasons for your answer.

(Exam continues next page ...)

QUESTION SIX

- (a) The rate at which a body cools in air is proportional to the difference between its temperature T and the constant temperature 20°C (in this case) of the surrounding air. This can be expressed by the differential equation:

$$\frac{dT}{dt} = -k(T - 20).$$

The original temperature of a heated metal bar was 100°C . The bar cools to 70°C in 10 minutes.

- (i) Show that $T = 20 + Ae^{-kt}$ is a solution to the differential equation.
- (ii) Show $A = 80$.
- (iii) Find the value of k .
- (iv) Find the time taken for the temperature of the bar to reach 60°C . (Give your answer to the nearest minute.)

(b) Suppose that $(5 + 2x)^{12} = \sum_{k=0}^{12} a_k x^k$.

- (i) Use the binomial theorem to write an expression for a_k .
- (ii) Show that $\frac{a_{k+1}}{a_k} = \frac{24 - 2k}{5k + 5}$.
- (c) Consider the geometric series $S = 1 + (1+x) + (1+x)^2 + \dots + (1+x)^n$.
- (i) Show that $S = \frac{(1+x)^{n+1} - 1}{x}$.
- (ii) Hence show that

$$S = {}^{n+1}C_1 + {}^{n+1}C_2 x + \dots + {}^{n+1}C_{r+1} x^r + \dots + {}^{n+1}C_{n+1} x^n.$$

- (iii) Hence prove

$${}^nC_r + {}^{n-1}C_r + {}^{n-2}C_r + \dots + {}^rC_r = {}^{n+1}C_{r+1}.$$

(Exam continues overleaf ...)

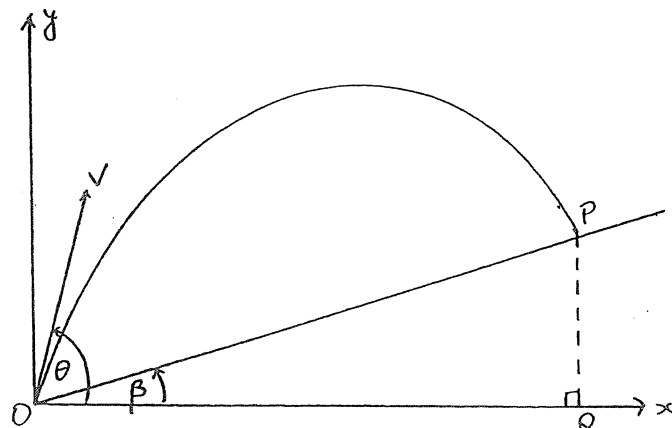
QUESTION SEVEN

(a) Consider the function $y = 2 \sin(x - \beta) \cos x$, where $0 < \beta < \frac{\pi}{2}$.

(i) Show that $\frac{dy}{dx} = 2 \cos(2x - \beta)$.

(ii) Hence or otherwise, show that $2 \sin(x - \beta) \cos x = \sin(2x - \beta) - \sin \beta$.

(b)



A projectile is fired from the origin with a velocity V and an angle of elevation θ , where $\theta \neq 90^\circ$. You may assume that:

$$x = Vt \cos \theta \quad \text{and} \quad y = -\frac{1}{2}gt^2 + Vt \sin \theta,$$

where x and y are the horizontal and vertical displacements of the projectile in metres from O at time t seconds after firing and g is the acceleration due to gravity.

(i) Show that the cartesian equation of the flight of the projectile is

$$y = x \tan \theta - \frac{g}{2V^2 \cos^2 \theta} x^2.$$

(ii) Suppose the projectile is fired up a plane inclined at β to the horizontal so that $0^\circ < \beta < \theta$. If the projectile strikes the plane at $P(h, k)$ show that

$$h = \frac{(\tan \theta - \tan \beta) 2V^2 \cos^2 \theta}{g}.$$

(iii) Hence show that the range OP of the projectile can be given by:

$$OP = \frac{2V^2 \sin(\theta - \beta) \cos \theta}{g \cos^2 \beta}.$$

(iv) By referring to (ii) of part (a) or otherwise, show that the maximum value of the range OP is given by $\frac{V^2}{g(1 + \sin \beta)}$.

(v) If the angle of inclination of the plane is 14° , at what angle to the horizontal should the projectile be fired in order to attain maximum range?

15 marks/question.

$$(a) (i) \int_0^1 \frac{x}{x^2+1} dx = \frac{1}{2} [\ln(x^2+1)]_0^1 \quad \checkmark$$

$$= \frac{1}{2} (\ln 2 - \ln 1)$$

$$= \frac{1}{2} \ln 2 \quad \checkmark$$

$$(ii) \int_{-2}^{2\sqrt{3}} \frac{1}{4+x^2} dx = \frac{1}{2} [\tan^{-1} \frac{x}{2}]_{-2}^{2\sqrt{3}} \quad \checkmark$$

$$= \frac{1}{2} (\tan^{-1} \sqrt{3} - \tan^{-1} -1) \quad \checkmark$$

$$= \frac{1}{2} (\frac{\pi}{3} + \frac{\pi}{4})$$

$$= \frac{7\pi}{24} \quad \checkmark$$

5m

$$(b) y = \tan^{-1}(\sin x)$$

$$\frac{dy}{dx} = \frac{\cos x}{1+\sin^2 x} \quad \checkmark$$

$$f'(0) = 1 \quad \checkmark$$

2m

$$(c) \frac{1}{x+1} < 3$$

$$x+1 < 3(x+1)^2 \quad \checkmark$$

$$3x^2 + 6x + 3 - x - 1 > 0$$

$$3x^2 + 5x + 2 > 0 \quad \checkmark$$

$$(3x+2)(x+1) > 0$$

$$x < -1 \text{ or } x > -\frac{2}{3} \quad \checkmark$$

3m

$$(d) \cos(\theta + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}$$

$$\theta + \frac{\pi}{4} = 2m\pi \pm \frac{\pi}{4} \quad \checkmark$$

$$\theta = 2m\pi \text{ or } 2m\pi - \frac{\pi}{2} \quad \checkmark$$

3m

$$(e) f: \text{ let } y = 8x^3$$

$$f^{-1}: \text{ then } x = 8y^3 \quad \checkmark$$

$$y^3 = \frac{x}{8}$$

$$y = \frac{1}{2} \sqrt[3]{x}$$

$$\text{or } f^{-1}(x) = \frac{1}{2} \sqrt[3]{x}$$

2m

(a) R.T.P. $\frac{\sin 2x}{1 + \cos 2x} = \tan x$

L.H.S. = $\frac{2 \sin x \cos x}{1 + 2 \cos^2 x - 1}$

= $\frac{2 \sin x \cos x}{2 \cos^2 x}$

= $\frac{\sin x}{\cos x}$

= $\tan x$

(b) let $f(x) = x^3 - 103x + 102.5$

$f'(x) = 3x - 103$

$f(1) = .5$

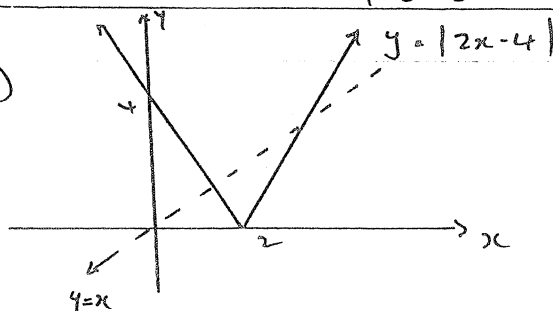
$f'(1) = -100$

$x_2 = 1 - \frac{f(1)}{f'(1)}$

= $1 - \frac{.5}{-100}$

= 1.005

(c)



for points of intersection

$x = 2x - 4$ or $x = -(2x - 4)$

$x = 4$

$x = \frac{4}{3}$

So, for $|2x - 4| > x$

$x < \frac{4}{3}$ or $x > 4$

(d)(i) $7 \cos \theta - \sin \theta = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$

$R \cos \alpha = 7$

$R \sin \alpha = 1$

$R = \sqrt{7^2 + 1^2} = 5\sqrt{2}$

$\alpha = \tan^{-1} \frac{1}{7}$

$\approx 8^\circ$

(ii) $7 \cos \theta - \sin \theta = 5$

$5\sqrt{2} \cos(\theta + \alpha) = 5$

$\cos(\theta + \alpha) = \frac{1}{\sqrt{2}}$

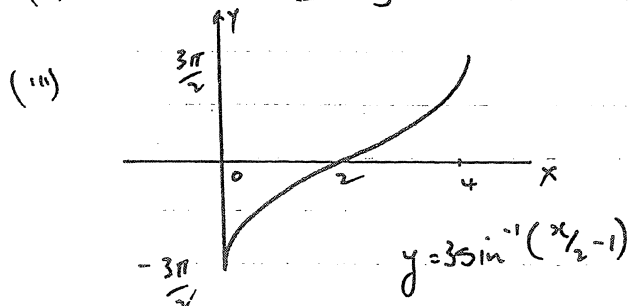
$\theta = 37^\circ, 308^\circ$

Q3

(a) (i) $D: -1 \leq \frac{x}{2} - 1 \leq 1$

$$0 \leq x \leq 4$$

(ii) $R: -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$



(iv) $f(1) = 3 \sin^{-1}(-\frac{1}{2})$
 $= -\frac{\pi}{2}$

5m

(b) $I = \int x \sqrt{1-x} dx$

$$u = 1-x$$

$$x = 1-u$$

$$\frac{du}{dx} = -1$$

$$I = \int (1-u) u^{1/2} - du$$

$$= \int u^{3/2} - u^{1/2} du$$

$$= \frac{2u^{5/2}}{5} - \frac{2u^{3/2}}{3} + C$$

$$= \frac{2}{5} \sqrt{(1-x)^5} - \frac{2}{3} \sqrt{(1-x)^3} + C$$

3m

(c) since $x^2 - 2x - 3 = (x-3)(x+1)$ and $(x^2 - 2x - 3)$ is a factor of $P(x)$ then $(x-3)$ & $(x+1)$ are factors of $P(x)$

$\therefore P(3) = 0$ i.e. $27 - 27 + 3a + b = 0$

$$3a + b = 0 \quad \text{--- (1)}$$

$P(-1) = 0$ i.e. $-1 - 3 - a + b = 0$

$$a - b = -4 \quad \text{--- (2)}$$

solve (1) & (2)

$$a = -1$$

$$b = 3$$

4m

(d) $\frac{d}{dx} \left(\tan^{-1} x + \tan^{-1} \frac{1}{x} \right) = \frac{1}{1+x^2} + \frac{-\frac{1}{x^2}}{1+\frac{1}{x^2}}$
 $= \frac{1}{1+x^2} - \frac{1}{1+x^2}$
 $= 0$

(ii) since $\frac{d}{dx} \left(\tan^{-1} x + \tan^{-1} \frac{1}{x} \right) = 0$

$$\tan^{-1} x + \tan^{-1} \frac{1}{x} = c$$

let $x=1$:

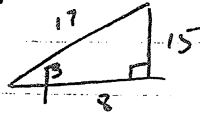
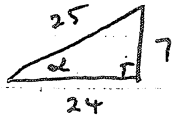
$$\tan^{-1} 1 + \tan^{-1} 1 = c$$

$$c = \pi/2$$

3m

Q4

1) let $\alpha = \cos^{-1} \frac{24}{25}$ & $\beta = \sin^{-1} \frac{15}{17}$



$$\begin{aligned}\cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta \\ &= \frac{24}{25} \cdot \frac{8}{17} + \frac{7}{25} \cdot \frac{15}{17} \\ &= \frac{295}{425}\end{aligned}$$

✓✓

} ✓

4 m

✓

(b) (i) $y = xe^{-x}$
 $\frac{dy}{dx} = e^{-x} - xe^{-x}$
 $= e^{-x}(1-x)$

for stationary points $\frac{dy}{dx} = 0$
 $e^{-x}(1-x) = 0$

$x = 1$

when $x = 1$, $y = e^{-1}$
 $\therefore A$ is $(1, \frac{1}{e})$

(ii) $R: y \leq \frac{1}{e}$

(iii) (a) 2

(b) 1

(c) 0

✓

✓

✓

✓

6 m

c (i) $\angle PTB = \angle TAB$ (alternate segment theorem)

(ii) $\angle TYX = \angle PTB + \angle TPY$ (exterior $\angle$ $\triangle TYP$)

$\angle TXY = \angle TAB + \angle XPA$ (ext. $\angle$ $\triangle PAX$)

Since $\angle PTB = \angle TAB$ & $\angle TPY = \angle APX$

$\angle TYX = \angle TXY$

So $TX = TY$ (isosceles $\triangle$)

(iii) $\triangle TYP \parallel \triangle APX$ (A.A.)

So $\frac{TY}{AX} = \frac{TP}{AP}$ (comp. sides of sim. $\triangle$'s in proportion)

$\therefore \frac{TX}{XA} = \frac{TP}{PA}$ (fr. (iii))

✓

} ✓✓

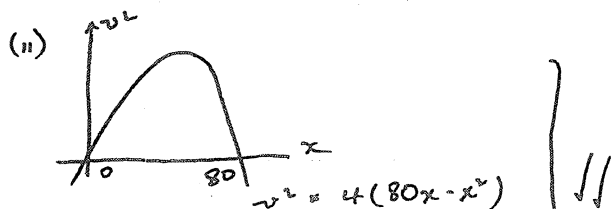
✓

} ✓

5 m

Q5

$$\begin{aligned}
 (i) \quad \ddot{x} &= 4(40-x) \\
 \frac{d}{dx}(\frac{1}{2}v^2) &= 4(40-x) \\
 \frac{1}{2}v^2 &= 4(40x - \frac{1}{2}x^2) + c \\
 \text{when } x=0, v=0 \therefore c=0 \\
 \frac{1}{2}v^2 &= 4(40x - \frac{1}{2}x^2) \\
 v^2 &= 4(80x - x^2)
 \end{aligned}$$



since $v^2 \geq 0$ for all x
 then v^2 is defined for $0 \leq x \leq 80$
 hence the particle moves in the
 interval $0 \leq x \leq 80$.

(iii) fr. graph max velocity
 when $x=40$
 i.e. $v^2 = 4(80 \times 40 - 40^2)$
 $= 80$.

max vel. 80 cm/s
 when $x = 40$ cm. 5m

(b) (i) $x^2 = 4ay$
 grad. of tangent at $P = p$
 grad. of normal at $P = -1/p$
 eqn. of normal at P :
 $y - ap^2 = -1/p(x - 2ap)$
 $py - ap^3 = -x + 2ap$
 $x + py = 2ap + ap^3$

(ii) for Q let $x=0$
 $py = 2ap + ap^3$
 $y = 2a + ap^2$
 hence Q is $(0, 2a + ap^2)$

(iii) $x = \frac{-mx_2 + m x_1}{-m + n}$

$$= \frac{0 + 1 \times 2ap}{-2 + 1}$$

$$= -2ap$$

$$y = \frac{-2(2a + ap^2) + ap^2}{-1}$$

$$= 4a + ap^2$$

$$\therefore R \text{ is } (-2ap, 4a + ap^2)$$

(iv) $x = -2ap$

$$p = \frac{-x}{2a}$$

$$y = 4a + ap^2$$

$$= 4a + a\left(\frac{x}{-2a}\right)^2$$

$$= 4a + \frac{x^2}{4a}$$

$$x^2 = 4a(y - 4a)$$

parabola

vertex $(0, 4a)$

focal length $= a$

(v) since $x + py = 2ap + ap^3$
 is through (h, k)

$$h + pk = 2ap + ap^3$$

$$ap^3 + (2a - k)p - h = 0$$

(vi) since $ap^3 + (2a - k)p - h = 0$

is a cubic equation in
 p the maximum number
 of solutions for p is 3
 hence the maximum
 number of points for P
 is 3.

10m

Q6

$$\begin{aligned} (i) \quad T &= 20 + Ae^{-kt} \\ \frac{dT}{dt} &= -kAe^{-kt} \\ &= -k(T-20) \end{aligned} \quad \left. \vphantom{\begin{aligned} T &= 20 + Ae^{-kt} \\ \frac{dT}{dt} &= -kAe^{-kt} \\ &= -k(T-20) \end{aligned}} \right\} \checkmark$$

$$\begin{aligned} (ii) \quad \text{when } t=0, T=100 \\ \text{So, } 100 &= 20 + Ae^0 \\ A &= 80 \end{aligned} \quad \left. \vphantom{\begin{aligned} 100 &= 20 + Ae^0 \\ A &= 80 \end{aligned}} \right\} \checkmark$$

$$\begin{aligned} (iii) \quad \text{when } t=10, T=70 \\ 70 &= 20 + 80e^{-10k} \\ e^{-10k} &= \frac{5}{8} \\ k &= -\frac{1}{10} \ln \frac{5}{8} \end{aligned} \quad \left. \vphantom{\begin{aligned} 70 &= 20 + 80e^{-10k} \\ e^{-10k} &= \frac{5}{8} \\ k &= -\frac{1}{10} \ln \frac{5}{8} \end{aligned}} \right\} \checkmark$$

$$\begin{aligned} (iv) \quad \text{when } T=60 \\ 60 &= 20 + 80e^{-kt} \\ e^{-kt} &= 0.5 \\ t &= \frac{10 \ln 0.5}{\ln \frac{5}{8}} \\ &\doteq 15 \text{ min.} \end{aligned} \quad \left. \vphantom{\begin{aligned} 60 &= 20 + 80e^{-kt} \\ e^{-kt} &= 0.5 \\ t &= \frac{10 \ln 0.5}{\ln \frac{5}{8}} \\ &\doteq 15 \text{ min.} \end{aligned}} \right\} \checkmark$$

6m

$$\begin{aligned} (b) \quad (i) \quad U_{k+1} &= {}^{12}C_k 5^{12-k} (2x)^k \\ \therefore a_k &= {}^{12}C_k 5^{12-k} 2^k \end{aligned} \quad \checkmark$$

$$(ii) \quad \frac{a_{k+1}}{a_k} = \frac{{}^{12}C_{k+1} 5^{11-k} 2^{k+1}}{{}^{12}C_k 5^{12-k} 2^k} \quad \checkmark$$

$$\begin{aligned} &= \frac{12!}{[12-(k+1)]! (k+1)!} \cdot \frac{2}{5} \\ &= \frac{12!}{(12-k)! k!} \cdot \frac{2}{5} \\ &= \frac{(12-k) 2}{(k+1) 5} \\ &= \frac{24-2k}{5k+5} \end{aligned} \quad \left. \vphantom{\begin{aligned} &= \frac{12!}{[12-(k+1)]! (k+1)!} \cdot \frac{2}{5} \\ &= \frac{12!}{(12-k)! k!} \cdot \frac{2}{5} \\ &= \frac{(12-k) 2}{(k+1) 5} \\ &= \frac{24-2k}{5k+5} \end{aligned}} \right\} \checkmark$$

3m

$$(c) \text{ (i) } S = 1 + (1+x) + (1+x)^2 + \dots + (1+x)^n \quad \text{--- (1)}$$

$$a = 1, \quad r = (1+x), \quad \text{no. terms} = n+1$$

$$\therefore S = \frac{a(r^{n+1} - 1)}{r - 1}$$

$$= \frac{1 \{ (1+x)^{n+1} - 1 \}}{(1+x) - 1}$$

$$= \frac{(1+x)^{n+1} - 1}{x}$$

$$(ii) S = \frac{1 + {}^{n+1}C_1 x + {}^{n+1}C_2 x^2 + \dots + {}^{n+1}C_{n+1} x^{n+1} - 1}{x} \quad \checkmark$$

$$= {}^{n+1}C_1 + {}^{n+1}C_2 x + \dots + {}^{n+1}C_{n+1} x^n \quad \text{--- (2)} \quad \checkmark$$

(iii) equate terms in x^r from (1) & (2) ✓

from (1) take co-efft of x^r from $(1+x)^1 + (1+x)^2 + \dots + (1+x)^n$ ✓

$$\text{i.e. } {}^1C_r + {}^{2+1}C_r + {}^{3+2}C_r + \dots + {}^nC_r$$

counting back this is equivalent to

$${}^nC_r + {}^{n-1}C_r + {}^{n-2}C_r + \dots + {}^rC_r$$

from (2) co-efft of x^r is ${}^{n+1}C_{r+1}$ ✓

$$\text{So } {}^{n+1}C_{r+1} = {}^nC_r + {}^{n-1}C_r + {}^{n-2}C_r + \dots + {}^rC_r$$

Q.E.D.

Q7

$$(a) \quad y = 2 \sin(x-\beta) \cos x$$

$$(i) \quad \frac{dy}{dx} = 2 [\cos(x-\beta) \cos x - \sin x \sin(x-\beta)]$$

$$= 2 \cos(x-\beta+x)$$

$$= 2 \cos(2x-\beta)$$

$$(ii) \quad \int 2 \cos(2x-\beta) dx = 2 \cdot \left[\frac{1}{2} \sin(2x-\beta) + C \right]$$

$$= \sin(2x-\beta) + K$$

$$\therefore 2 \sin(x-\beta) \cos x = \sin(2x-\beta) + K$$

$$\text{let } x=0: \quad 2 \sin(-\beta) = \sin(-\beta) + K$$

$$K = \sin(-\beta)$$

$$= -\sin \beta$$

$$\text{So, } 2 \sin(x-\beta) \cos x = \sin(2x-\beta) - \sin \beta$$

$$(b) (i) \quad t = \frac{x}{v \cos \theta}$$

$$\therefore y = -\frac{1}{2} g \left(\frac{x}{v \cos \theta} \right)^2 + v \left(\frac{x}{v \cos \theta} \right) \sin \theta$$

$$= x \tan \theta - \frac{g}{2v^2 \cos^2 \theta} x^2 \quad \text{--- (A)}$$

$$(ii) \quad \text{for P: solve (A) \& } y = x \tan \beta$$

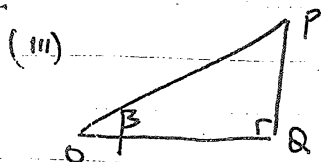
$$\therefore x \tan \beta = x \tan \theta - \frac{g}{2v^2 \cos^2 \theta} x^2$$

$$\frac{g}{2v^2 \cos^2 \theta} x^2 + x (\tan \beta - \tan \theta) = 0$$

$$x \left[\frac{gx}{2v^2 \cos^2 \theta} - (\tan \theta - \tan \beta) \right] = 0$$

$$x=0 \quad \text{or} \quad \frac{(\tan \theta - \tan \beta) 2v^2 \cos^2 \theta}{g}$$

$$\text{So, } h = \frac{(\tan \theta - \tan \beta) 2v^2 \cos^2 \theta}{g}$$



$$OP = \frac{OQ}{\cos \beta} = \frac{h}{\cos \beta}$$

$$\text{So, } OP = \frac{2v^2}{g} \left(\frac{\sin \theta}{\cos \theta} - \frac{\sin \beta}{\cos \beta} \right) \frac{\cos^2 \theta}{\cos \beta}$$

$$= \frac{2v^2}{g} \left(\frac{\sin \theta \cdot \cos \beta - \cos \theta \sin \beta}{\cos \theta \cdot \cos^2 \beta} \right) \cos^2 \theta$$

$$= \frac{2v^2}{g} \frac{\sin(\theta - \beta) \cos \theta}{\cos^2 \beta}$$

(iv) from (a) (ii)

$$OP = \frac{2v^2 \sin(\theta - \beta) \cos \theta}{g \cos^2 \beta}$$

$$= \frac{v^2 [\sin(2\theta - \beta) - \sin \beta]}{g \cos^2 \beta}$$

for maximum $\sin(2\theta - \beta) = 1$

$$\text{So } OP = \frac{v^2 [1 - \sin \beta]}{g \cos^2 \beta}$$

$$= \frac{v^2 (1 - \sin \beta)}{g (1 - \sin^2 \beta)}$$

$$= \frac{v^2}{g(1 + \sin \beta)}$$

(v) for max. range $2\theta - \beta = 90^\circ$

$$2\theta = (90^\circ + 14^\circ)$$

$$\theta = 52^\circ$$

